

Using Fuzzy Techniques in Business Performance Management

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Abstract—An important part of Business Performance Management (BPM) is identifying strategic targets for the organisation, cascading them down into tactical and operational targets and finally optimising operational levers or actions as to achieve the given targets. In order to do this, performance metrics have to be defined at all levels and relationships between levers and metrics have to be mathematically modelled. Thereby, traditional BPM cannot deal with imprecise information although imprecision is a commonplace in real world applications. We discuss the most important functions of BPM and show how fuzzy and probabilistic techniques can be applied to process imprecise information. A few of the techniques have already been implemented in a research prototype.

I. INTRODUCTION

With growing competitiveness in the global market place it is paramount for any organisation to carefully manage its performance in order to be successful. Bottom line of performance management is to take the right decisions at the right time such that strategic targets can be achieved. Popular tools include Balanced Scorecards [1] and Strategy Maps [2] for strategic planning and Business Intelligence platforms for reporting current performance, comparing it to set targets and raising alerts if necessary.

Balanced Scorecards and Strategy Maps usually describe business strategies at quite a high level like 'strengthen portfolio and improve advertisement in order to increase sales and finally profit'. Such strategies have to be translated into more concrete actions and business processes at all levels of an organisation (strategic, tactical and operational). Performance metrics have to be defined that on the one hand measure success at all levels and on the other hand drive success of the strategy. In other words, if the value of a metric is not correlated with fulfilling the strategy, then it should be dropped. Business Intelligence platforms plug into operational and tactical data sources and provide a view onto business performance thereby comparing actual values of performance metrics with set targets.

Business users are then keen to know what they can change in the business (operational levers) or what might happen to external influences in the future and how such changes would affect overall business performance. Playing with such scenarios is called *what-if analysis*. It requires knowledge of how different performance metrics are mathematically related. Such functional relationships are either known to domain

experts and can even be quite simple at times, or they can be learned from historic data using machine learning techniques.

Furthermore, business users set targets for strategic performance metrics and then need to know how such high-level targets translate into lower-level targets (tactical and operational) and how operational levers have to be set to achieve high level targets. It turns out that this is an optimisation problem we will describe in greater detail in Section IV.

Up to here we assumed a deterministic setting with precise information. All metrics produce crisp values. However, in most of our applications this is not the case and not a reasonable simplification either. In a typical scenario at least the values of external influences are imprecise, in particular, if we consider predicted future values. For instance, we might want to predict the number of orders we will receive from our customers on the following day and see what this means for metrics like customer satisfaction. The prediction model for orders would be based on historic data in this case and the predicted value usually be a probability distribution or histogram. For what-if analysis, such uncertain values need to be propagated to all metrics that are directly or indirectly linked to the uncertain one.

The optimisation of operational levers turns out to be difficult in case of imprecise values, since we hit the classic problem of fuzzy optimisation, i.e. we have to deal with imprecise values of the objective function.

After the introduction of *Business Performance Frameworks* as main instrument of performance management in Section II, we discuss different approaches of propagating imprecise information in Section III. Optimisation of business targets and operational levers given imprecise values of performance metrics is introduced in Section IV. The paper closes with experimental results in Section V that show the advantage of using imprecise information. In all sections, different solutions are compared in terms of their usefulness for business users.

II. BUSINESS PERFORMANCE FRAMEWORKS

Main functions in Business Performance Management are setting targets, monitoring performance against the targets, making performance predictions and optimising operational levers and business processes accordingly as to improve overall performance. As a foundation for all these functions we take the following steps.

- 1) decide on factors like *profit* that are important for the strategy of the organisation
- 2) figure out at tactical and operational levels what factors might contribute to strategic factors
- 3) determine operational levers (what can be controlled within the organisation) and external influences (cannot be controlled)
- 4) decide how to measure the performance for each factor (definition of metrics)
- 5) model relationships in a directed graph between levers and factors

The result of this exercise is a directed graph with functional relationships in the vertices. We call such a graph *business performance framework (BPF)*. A simple example related to customer contact centres and fault resolution is shown in Fig. 1. Strategic factors in this case are cost and customer satisfaction. They are driven by the time to clear a fault (customers might report faults), the time a customer waits in a calling queue and the number of abandoned calls. More human related, soft factors are staff motivation and competency. Operational levers are the numbers of technicians who can resolve faults and the number of contact centre agents as well as the number of training courses and team-building events the staff attend. Call rate measured in number of customer calls per minute is the external influence. The values next to the boxes represent the current value, however they are only fictitious for demonstration purposes. The links indicate the functional relationships, e.g. $\text{cleartime} = f(\text{competency}, \text{technicians}, \text{jobrate})$.

Monitoring performance is quite straightforward in so far as metric definitions are applied to data sources and the results are shown in reports or real-time dashboards. Merely data quality issues and the fact that data is typically distributed over a variety of different sources complicate this task in practice.

Finding targets for an entire organisation given strategic goals as well as optimising operational parameters however is much more difficult, the main reason being that an understanding of how the business works needs to be developed which is done in the last point in the list above.

Due to the functional relationships, values of metrics can be propagated from the *leaf nodes* (the ones with no incoming edges) through the graph up to the *root nodes* (the ones with no outgoing edges). This function is called *what-if analysis* since an analyst can set values for the leaf nodes, i.e. create a business scenario, and then see the resulting values for all the other metrics. For instance, if we increased the number of technicians, we would expect clear time to drop and customer satisfaction to increase. At the same time, cost would go up.

Some of the relationships might be dynamic, i.e. they have to be modelled as a differential equations rather than functions. Since the framework only allows us to define functional relationships, nodes can instead keep a history of previous values and use these as inputs for the relationship function. Many time series prediction algorithms make use of this trick which is based on the idea that the derivatives in differential equations can be approximated by functions of previous values. A very simple example is the length of a calling queue that can be

computed as previous length plus number of received calls minus number of calls taken.

The propagation mechanism of what-if analysis can also be used to predict future values of metrics higher up in the hierarchy. This can be achieved by predicting the future values of external influences (these are leaf nodes) and propagating these through the graph leading to predicted values for the remaining metrics. For instance, we could predict the volume of incoming orders (external influence) for the following day and in combination with the planned deployment in customer contact centres (operational lever) determine customer satisfaction for the following day.

What-if analysis helps understand an organisation and can also be used for planning purposes. However, business users finally want to take a top-down approach. They would like to set strategic targets and then need to know, how operational levers have to be set such that the strategic targets will be achieved. The functional relationships all point upwards in the hierarchy from operational levers and external influences to operational to tactical and finally to strategic metrics. Such relationships are not one-to-one mappings – they are often neither surjective nor injective, i.e. they cannot be inversed. In other words, target values cannot simply be propagated down from strategic factors to operational levers. In fact, strategic targets might be achievable by a number of different value combinations in the leaf nodes if a solution exists at all. Managers will be interested in the best solution, which turns the task into an optimisation problem. An objective function representing quantities like cost or utility has to be defined and typically a heuristic search algorithm employed to find the solution that minimises or maximises the objective function.

III. PROPAGATING IMPRECISE INFORMATION

A lot of the information and knowledge business performance management is concerned with is imprecise. As usual, we can distinguish uncertain and vague information. Typical causes of uncertainty are the aggregation of information over time, business processes being executed inconsistently or simply the lack of complete knowledge. An example for the last type is a predicted future value of an external influence. Anything related to customer behaviour, for instance, is difficult to predict with certainty.

Vagueness comes into play when information is the result of human perception. Relationships between performance metrics can sometimes be best described as a set of fuzzy rules and values for leaf nodes in what-if analysis are often appropriately represented as fuzzy concepts.

In this paper, we only deal with precise relationships between external influences, operational levers and performance metrics. We therefore assume that the relationships are represented as crisp functions, but that the value of a node can be imprecise. Consequently we require a mechanism to propagate imprecise information through the graph of a BPF.

The standard approach for uncertainty propagation is Monte Carlo simulation. For each leaf node a random value will be drawn from an assumed probability distribution and the

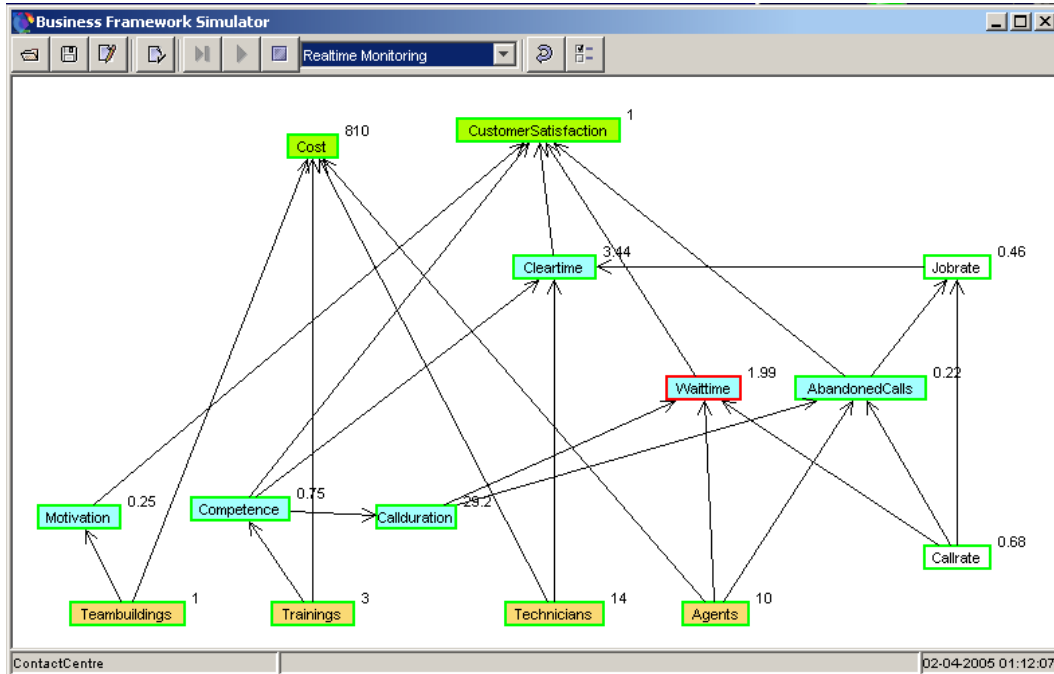


Fig. 1. Example of a Business Performance Framework (BPF)

values will be propagated through the graph. This process will be repeated n times and in this way a histogram can be constructed for each node. In order to reduce the number of repetitions, techniques like Latin Hypercube Sampling [3] can be employed, where the universes of discourse are divided into m regions of equal probability mass and random values are drawn from each region in turn. In this way, the ranges of likely values can be covered more evenly with fewer repetitions.

If the values of the leaf nodes are fuzzy sets, Zadeh's extension principle [4] can be applied for the propagation of fuzzy membership values. In theory, all possible value combinations of leaf nodes have to be taken into account which is usually not feasible due to exponential computational complexity. In practice, sampling techniques like discretisation or parametric models can be applied to solve the problem.

Many business users struggle to understand probability distributions and fuzzy sets to such an extent that they cannot use them to make business decisions. Having said that predefined fuzzy sets based on vague concepts like 'high' customer satisfaction are comprehensible, but fuzzy sets resulting from value propagation based on the extension principle are usually not interpretable in such terms anymore. Such users are however happy to deal with set-valued metrics. For instance, an interval as range of likely values of a metric is perfectly acceptable.

As crisp sets are special cases of probability distributions and fuzzy sets, the same value propagation techniques are applicable. Furthermore, given certain assumptions regarding the functional relationships, computational complexity can be reduced considerably. For instance, if the set is a closed

interval and the relationship function is monotonous then only the upper and lower boundary of the interval have to be propagated. The two resulting values then form the boundaries of the result set.

A useful extension to set-valued metrics are confidence levels which again are accepted in the business user community. Rather than showing fuzzy sets or probability distributions, a representation as a set of α -cuts is therefore a good alternative. The user can select a value for α as confidence level and the corresponding cut is shown as a set of values.

IV. FUZZY OPTIMISATION OF BUSINESS TARGETS

Implementing the strategy of an organisation involves setting targets at all levels and finding the right values for operational levers and process parameters. In practice, high level managers will set target values for the strategic metrics, i.e. the root nodes of a BPF. A target value is usually defined as a threshold that separates desired from non-desired values. More generally, a target is a set of values forming a constraint on acceptable values for the respective performance metric. An example is "customer satisfaction larger than 95%". The values of the external influences have to be set as well. These represent the assumptions the user makes about the state of the business environment.

The problem then is to find values for the operational levers, which are all located in the leaf nodes of the BPF. These values must be chosen in such a way that when propagated up through the BPF, the resulting values for the strategic root nodes fulfil the defined constraints. If a solution for the leaf values exists, it is usually not unique. In order to pick a solution for the user, we therefore employ objective functions (cost or utility) and

turn the task into an optimisation problem. A simple example is “achieve customer satisfaction larger than 95% at minimal cost”.

For optimisation we use the iOpt toolkit providing heuristic search-based solutions. It allows the user to state his problem as a Constrained Optimisation Problem (COP) using decision variables, constraints and objective function, and to easily implement metaheuristics such as single solution algorithms (Local Search, Tabu Search, Simulated Annealing), population-based algorithms (such as Genetic Algorithms), and hybrid methods being a combination of the two [5], [6].

In many of our application scenarios, external influences in particular introduce imprecision into BPFs. As external influences represent the usually dynamic business environment, assuming a crisp value would not be a reasonable simplification. Consequently, the values of all the root nodes directly or indirectly connected to an imprecise node will be imprecise, too. So will be the value of the objective function, since the values of the root nodes typically feed into it. Standard optimisation procedures cannot cope with imprecise values of the objective function, because it is difficult to rank solutions. The reason is that there is no natural order relation for imprecise values. Recent results in dealing with fuzzy parameters in objective functions can be found in [7].

Fortunately, the area of risk management provides a solution to the problem. As many other researchers we advocate to combine performance and risk management, the reason being that an organisation should primarily be interested in performance maximisation rather than risk minimisation. Threats, lack of knowledge and other factors introduce imprecision mostly in form of uncertainty into BPFs. The result are risks that are attached to metrics in a BPF like the “risk of losses”. Now, given an imprecise value of a metric in form of a probability distribution or fuzzy set, the question arises how to assess risk. A common assessment is called *value at risk*, one variety of which can be defined as the expected crisp value in the $n\%$ worst cases. For instance, if we consider a probability distribution for the metric “losses” we could compute the expected value of the top 10% of the probability mass. An alternative is the value x such that $P(\text{losses} > x) = n\%$, i.e. the chance of suffering losses larger than x is $n\%$.

Rather than using imprecise values of metrics in the objective function we can therefore map an imprecise value onto a crisp one using functions like value at risk. In that way, normal optimisation techniques can be applied. Only the propagation of imprecise values in BPFs has to be embedded.

V. EXPERIMENTS

In this section we will discuss practical results we gained from a series of experiments. They are related to real applications in our organisation and have been conducted with our advanced research prototype of a performance management platform.

In particular, we made experiments with what-if analysis and the optimisation of lever values given imprecise influences. The results show that taking imprecision explicitly into

account delivers more information, since solutions convey the degree of their imprecision rather than just being a single value. Furthermore, some solutions reveal that alternative results can be almost as likely or possible, although they might be quite different. In fact, in a few cases a result will be obtained that is considerably different from the crisp solution because imprecision had been taken into account.

For the experiments we used the framework shown in Fig. 2. This example is related to flexible ICT infrastructures, where the size and parameters of a server farm can be adjusted on the fly to cater for a given demand. The levers are number of servers, bandwidth and page size (different sizes of web pages by, for instance, varying the resolution of pictures). Requests hitting the server farm are the external influence, response time, timeouts and transfer time are tactical performance indicators and finally dissatisfaction and costs are strategic performance measures. Dissatisfaction measures the time it takes an end user to open up a web page (measured in tenth of a second, or short ds for deciseconds). The relationships between the measures have been handcrafted using pipeline models.

In this more technical problem compared to the service scenario in Fig. 1, we can see quite clearly that we are dealing with a complex closed-loop control problem. Given targets for dissatisfaction and costs, the three levers have to be optimised following the values of the external influence. The objective functions we employed are weighted sums of dissatisfaction and costs and have to be minimised as to potentially minimise both factors.

Fig. 3 shows a value for the strategic performance measure dissatisfaction as the result of a what-if analysis. We assumed a triangular membership function for the number of requests and also a fuzzy value for the number of servers based on the likelihood of server failure.

We can clearly see two peaks, one around a value of 27ds, the second one in the region of 36-42ds. Furthermore, values in the range of 13-57ds carry a considerable degree of membership. Executed in crisp mode (peaks of the fuzzy levers as their crisp values) the framework produced 27ds which is consistent with the highest peak. Looking at the quite high level of imprecision of the fuzzy result, i.e. considering sources of imprecision in performance models it becomes clear that such a single value is not very trustworthy.

Optimising the lever values basically means varying their values as to minimise the objective function. If we compare the results for dissatisfaction in Figures 3 and 4, we can see that the defuzzified values are almost identical, however the tails of the fuzzy sets are considerably different. Thereby we looked for the maximum membership value closest to the centre of gravity as defuzzification mechanism. As we have to use defuzzified values in the objective function the question arises if the chosen defuzzification method is appropriate since it does not fully take the tail into account. As an alternative, we used the second variant of value at risk at confidence level 90% as described in Section IV. In the first setting, value at risk is about 689 seconds with costs of 10174, in

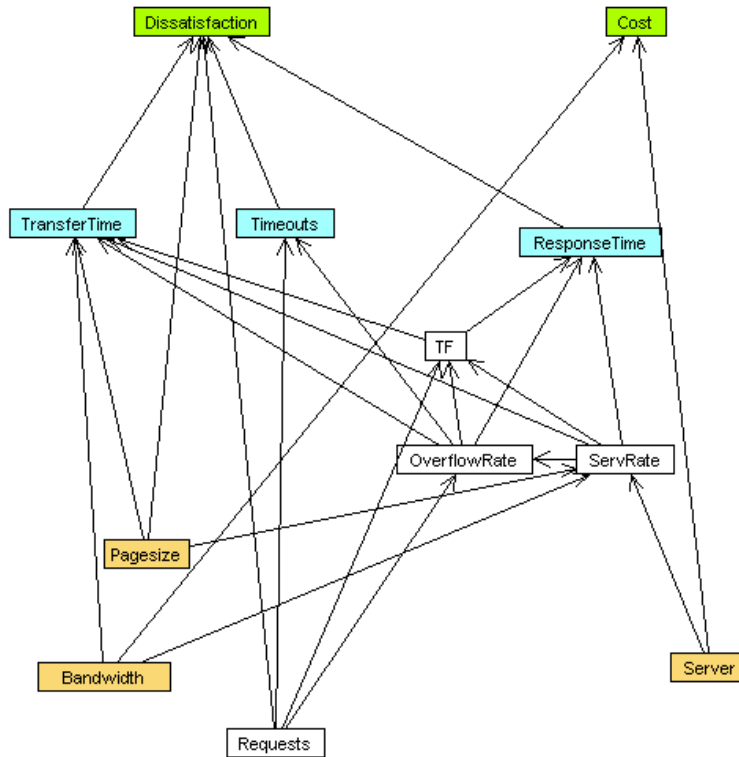


Fig. 2. Performance framework for a server farm hosting a service. Dissatisfaction is the time (in tenth of a second) in takes to open a web page.

the second one about 429 seconds at costs of 11174. Where the straightforward optimisation procedure would just prefer the first configuration due to lower costs at almost identical expected dissatisfaction, the second configuration would be picked if value at risk were to be taken into account in the objective function with an appropriate weight. We can clearly see that explicit modeling of imprecision introduces more flexibility into the optimisation procedure. Objectives can be specified based on a greater variety of factors eventually leading to potentially more reliable results.

VI. CONCLUSION

Our research shows that current performance management systems cannot deal with imprecise information. They assume crisp deterministic information thereby ignoring situations that might be as likely or possible as the considered one. This fact becomes even more apparent when risk has to be taken into account as it is mandatory with the introduction of regulations like Basel II and Sarbanes Oxley. Performance and risk management should be combined in a single framework where an organisation should endeavour to maximise its performance under risk. Thereby, risk is introduced by threats that are inherently linked to imprecision. We might not know exactly when a threat becomes active nor might we know its precise impact.

We have reviewed the most important functions of performance management and discussed different solutions to handle

imprecision. Our experiments show that explicitly dealing with imprecision gives more informed results and greater flexibility for the definition of performance objectives as it might be crucial for decision making.

Future work is concerned with extending the framework to allow for imprecise relationships like fuzzy functions or continuous Bayesian networks.

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Fig. 3. Fuzzy value for dissatisfaction as the result of what-if analysis with imprecise lever values (dissatisfaction on the x-axis in deciseconds and degree of membership on the y-axis). Defuzzified value is 28.0856 deciseconds and the value for costs is 10174.

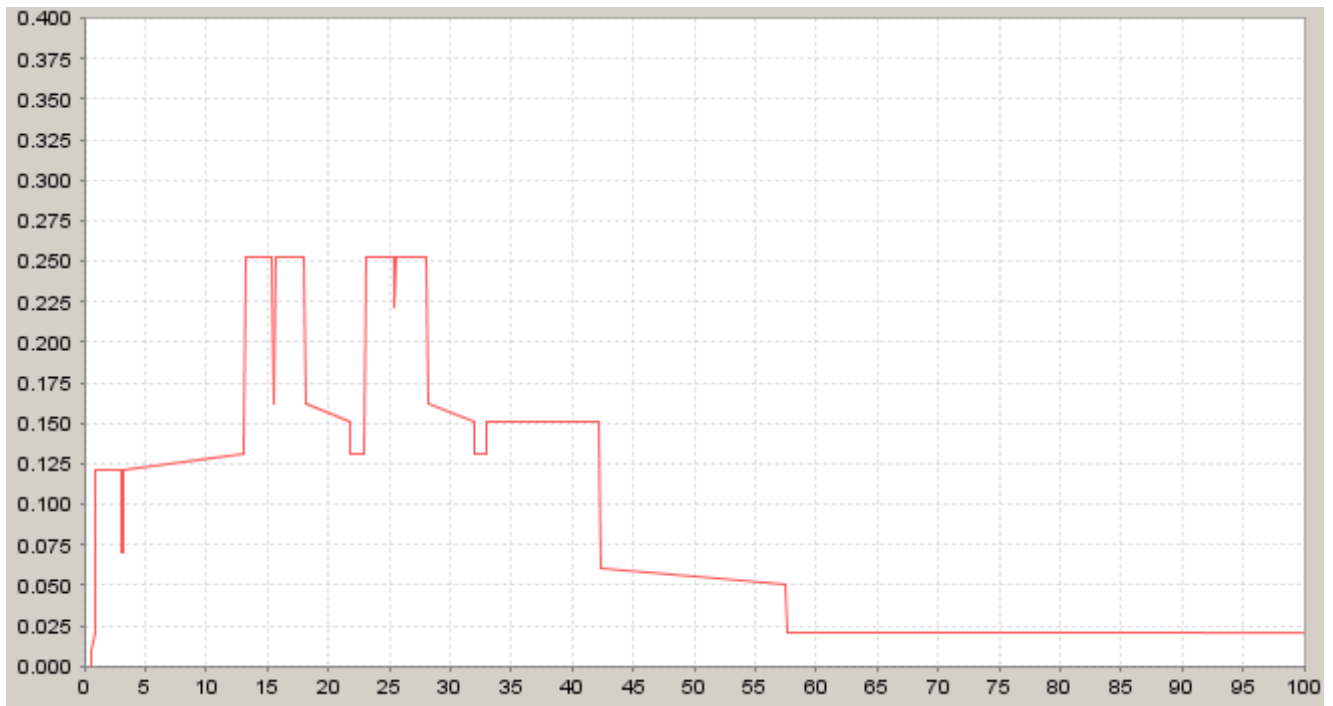


Fig. 4. Fuzzy value for dissatisfaction as the result of what-if analysis with imprecise lever values (dissatisfaction on the x-axis in deciseconds and degree of membership on the y-axis). Defuzzified value is 28.1103 deciseconds and the value for costs is 11174.